Each row of the matrix is labeled with a single index ($a'b'$) of the composite system and each column with (ab).

The vectors $|ab\rangle$ are taken to be orthonormal, which means that their inner products are zero unless both labels match. This does *not* mean that a matches b, but rather that ab matches $a'b'$. We can also express this idea using the Kronecker delta symbol:

$$\langle ab|a'b'\rangle \;=\; \delta_{aa'}\delta_{bb'}.$$

The right side is zero unless $a = a'$ and $b = b'$. If the labels do match, the inner product is one.

Now that we have the basis vectors, any linear superposition of them is allowed. Thus, any state in the composite system can be expanded as

$$|\Psi\rangle = \sum_{a,b} \psi(a,b)|ab\rangle.$$

6.4 TWO SPINS

Returning to our example, let's imagine two spins: Alice's and Bob's. To put it in a context that we can visualize, imagine that the spins are attached to two particles and that the two particles are fixed in space at two nearby but different locations.

Alice and Bob each have their own apparatuses, called $\mathcal{A}$ and $\mathcal{B}$ respectively, that they can use to prepare states and measure spin components. Each can be independently oriented along any axis.

We are going to need names for the two spins. When we only had one spin, we simply called it σ, and it had three components along the x,y, and z axes. Now we have two spins, and the question is how to label them without cluttering the symbols with too many sub-and superscripts. We could call them σ^A and σ^B, and the components, σ_x^A,σ_y^B, and so on. For me, that's just too many subscripts to keep track of, especially on the blackboard. Instead, I'll follow the same convention we used in the Interlude on tensor products.